Original research

Bacterial genomic structural variations in children with autism serve as diagnostic biomarkers

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ABSTRACT

Background Gut microbiota dysbiosis is linked to autism spectrum disorder (ASD) in children. However, the role of bacterial genomic structural variations (SVs) in ASD remains largely unexplored.

Objective We aimed to identify bacterial SVs associated with ASD and explore their mechanistic role and clinical application.

Design We collected faecal metagenomes from 452 children (261 ASD, 191 neurotypical) across an in-house and seven public datasets. Using linear mixed-effects modelling, we identified ASD-associated SVs and compositional shifts and validated candidate SVs in humanised gut microbiome mice.

Results We identified 100 bacterial SVs significantly associated with ASD ($p<0.05$). These SVs were enriched in genes involved in critical biological processes, including ion and amino acid metabolism and bacterial growth regulation in ASD. In particular, we found important SVs in *Bacteroides uniformis* related to thiamine and iron metabolism. Moreover, SVs in *Ruminococcus torques* were associated with the *MazF* (endoribonuclease toxin) and *MazE* (antitoxin) system, a key regulator of pathobiont proliferation. Validation in humanised mouse models confirmed significant correlations between these SV signatures and ASD-like behaviours, such as reduced social interaction and increased repetitive behaviours. Both phylogeographically conserved and regionally restricted SVs showed strong associations with ASD. A diagnostic model combining nine SVs and three bacterial species achieved an area under the receiver operating characteristic curve of 81.1%, outperforming models based solely on variable SVs (79.1%), deletion SVs (75.2%) or bacterial species abundance alone (72.3%).

Conclusion Our findings suggest the significant role of bacterial genomic SVs in ASD and highlight their potential as diagnostic biomarkers.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The gut microbiome plays a critical role in children with autism spectrum disorders (ASD).

⇒ Bacterial genomic structural variations (SVs) can impact host phenotypes; however, their contribution to ASD remains uncharacterised.

WHAT THIS STUDY ADDS

⇒ We identified 100 bacterial SVs significantly associated with ASD ($p<0.05$) through multicohort analysis.

⇒ Distinct bacterial SV signatures were associated with ASD, and these SVs were functionally associated with metabolic dysregulation and gut dysbiosis commonly observed in this disorder.

⇒ Critical SVs in *Bacteroides uniformis* were shown to regulate neurodevelopment-related thiamine and iron metabolism, correlating with ASD-related behaviours, and were validated in a humanised mouse model.

⇒ SVs related to the toxin-antitoxin system in *Ruminococcus torques* were implicated in its pathological overgrowth in ASD.

⇒ A diagnostic panel combining nine SVs and three bacterial species achieved an area under the receiver operating characteristic curve of 81.1%, which was superior to the panel using bacterial species alone.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Gut bacterial genomic SVs are linked to gut metabolic dysregulation and dysbiosis observed in children with ASD.

⇒ This study reveals new potential for microbial genomic SV-based biomarkers in diagnosing childhood ASD.

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INTRODUCTION

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterised by persistent difficulties in social communication and interaction, along with restricted and repetitive behaviours.[1] The dramatic rise in the prevalence of ASD in children has emerged as a major public health challenge,[2] while the underlying pathogenesis remains poorly understood. The aetiology of ASD is multifactorial, involving a complex interplay of intrinsic genetic predispositions and extrinsic factors like gut dysbiosis.[1 3] Notably, studies have identified differences in microbial diversity and composition in children with ASD compared with neurotypical children,[4 5] with research identifying bacterial markers (eg, *Bacteroides* spp) showing promise for ASD diagnosis.[6 7] Such dysbiosis in the gut can disrupt the gut-brain axis, leading to the metabolic imbalances and neuroimmune dysregulation in the brain, which may contribute to ASD symptom pathogenesis.[8]

ASD is highly heritable, with converging evidence from twin and family studies indicating a substantial contribution of host genetic alterations to its development across diverse populations.[9] These genetic factors interact with environmental and biological modifiers to shape neurodevelopmental trajectories in individuals with ASD. Likewise, genomic structural variations (SVs) in bacteria can influence host phenotypes through gene gain, loss or rearrangement, in ways that extend beyond what can be inferred from microbial abundance profiles alone.[10] For instance, SVs have been genetically linked to host metabolic traits such as HbA1c (hemoglobin A1c) and total cholesterol levels[10] and to the production of specific metabolites including bile acids,[11] thereby providing direct evidence of functional microbial impact. Additionally, microbial SVs have also been associated with therapeutic metabolite and response changes during immune checkpoint blockade in cancer,[12] highlighting their relevance to treatment outcomes. These findings underscore how SVs, by altering bacterial gene function, can drive disease pathogenesis through changes in community structure, metabolomic output and immune modulation.[13] However, the association of gut bacterial SVs with ASD pathogenesis is unclear.

In this study, we analysed stool samples from 452 children globally (261 with ASD and 191 neurotypical controls), including newly collected samples from 64 ASD cases and 64 matched controls, via shotgun metagenomic sequencing. For the first time, we found that bacterial genomic SVs were significantly associated with ASD compared with neurotypical children, showing that key SVs within *Ruminococcus torques* and *Bacteroides uniformis* were linked to ASD. These ASD-associated SVs are involved in diverse biological functions, such as ion/amino acid metabolism and bacterial growth regulation, which align with the observed metabolic dysregulation and gut dysbiosis in ASD. For example, we identified an ASD-depleted SV in the 1689–1708 kb genomic region of *B. uniformis*, which encompasses genes critical for regulating thiamine and iron metabolism, a key pathway influencing neurodevelopment and behaviour.[14] The ASD-associated SVs identified in children were further validated using a humanised mouse model. Finally, we demonstrated that a diagnostic panel combining bacterial SVs and abundance markers achieved superior accuracy in distinguishing children with ASD from neurotypical children.

RESULTS

Gut bacteria genomic SVs in bacterial genome are associated with ASD

In this study, a total of 452 children (ASD=261 (age median=5; male/female=6.1) and neurotypical children=191 (age median=5; male/female=4.12)) with shotgun metagenomic sequencing data were analysed (figure 1A and online supplemental table S1). We first investigated the gut bacterial composition in the children with ASD and neurotypical children. We found that the gut alpha diversity in ASD was significantly reduced compared with that in neurotypical children ($p=0.017$, two-tailed Wilcoxon Rank Sum test) (figure 1B). Compared with neurotypical children, correspondence analysis revealed an altered bacterial composition in ASD ($p=0.034$, permutational multivariate analysis of variance (PERMANOVA) adjusted for cohort effect (figure 1C)). To ensure that the identified differences were not driven by batch effects or demographic confounders, we performed stratified analyses by key variables, including geography, sex and individual cohort. These analyses showed that the differences in bacterial composition between ASD and typically developing (TD) children were not confounded by geographic origin (online supplemental figure S1A) and sex (online supplemental figure S1B). To identify ASD-associated bacteria, we performed a differential analysis considering the bacterial prevalence, abundance and cohort consistency (online supplemental figure S1C–E and online supplemental table S2). We found that 47 gut bacterial species were significantly enriched in ASD compared with neurotypical children, including those reported ASD-associated bacteria *Bacteroides* spp (*Ba. fragilis*),[15] *Clostridium bolteae*[16] and *R. torques*[17]); while 46 species were depleted in ASD, including the beneficial bacteria *Bifidobacterium longum* and *Faecalibacterium prausnitzii* (online supplemental file figure S1F).

We then assessed the bacterial genomic SVs in children with ASD and neurotypical children. In total, we identified 140 SV-containing bacterial taxa out of 1541 detectable bacteria, with prevalence rates ranging from 0.21% to 80.53% across all samples (median=6.15% (Q1=2.05%; Q3=24.59%)). After filtering out SV-containing bacteria with prevalence <25%, the remaining 33 high-prevalent bacteria (eg, *Ba. uniformis* and *R. torques*) exhibited significant genomic alterations (number of SVs: median=146 (Q1=104; Q3=191)) in children (figure 2A). Among all SVs, 31.2% (n=4533) were identified as variable SVs (v-SVs; microbial genomic sequence showing variable coverage), while 68.8% (n=10 015) were deletion SVs (d-SVs; microbial genomic sequence with absent (near-zero coverage) in 25–75% of samples) (figure 2A). To investigate differences in bacterial SV profiles between ASD and neurotypical children, we analysed the dissimilarity of SV profiles using intersample dissimilarity metrics. Correspondence analysis demonstrated a significant difference in bacterial SV profile between ASD and neurotypical children ($p<0.05$, ANOVA like permutation test) (figure 2B; online supplemental figure S2A). We further evaluated whether SVs were influenced by potential confounders, showing that the difference in SV profiles between ASD and TD was not confounded by geographic origin (online supplemental figure S2B, C) and sex (online supplemental figure S2D), indicating that bacterial SVs are significantly associated with ASD status.

ASD-associated bacterial SVs are linked to diverse biological functions

To identify differentially distributed SVs between ASD and neurotypical children, we applied a linear mixed-effects model adjusted for age and sex. Our analysis revealed 100 statistically significant SVs, comprising 26 v-SVs (figure 2C and online supplemental table S3) and 74 d-SVs (figure 2D and online supplemental table S4), which showed differential distribution between ASD and neurotypical children. Among these SVs, we observed significant enrichment of seven v-SVs and 31 d-SVs in ASD compared with neurotypical children, while 19 v-SVs and 43 d-SVs were enriched in neurotypical children. These SVs were distributed across 29 bacterial species, including *Ba. uniformis* and *R. torques* (figure 2C, D). Subsequent functional annotations of these SVs revealed their association with diverse biological processes (figure 2C, D), including ion/amino acid metabolism (which plays a critical role in maintaining neurological

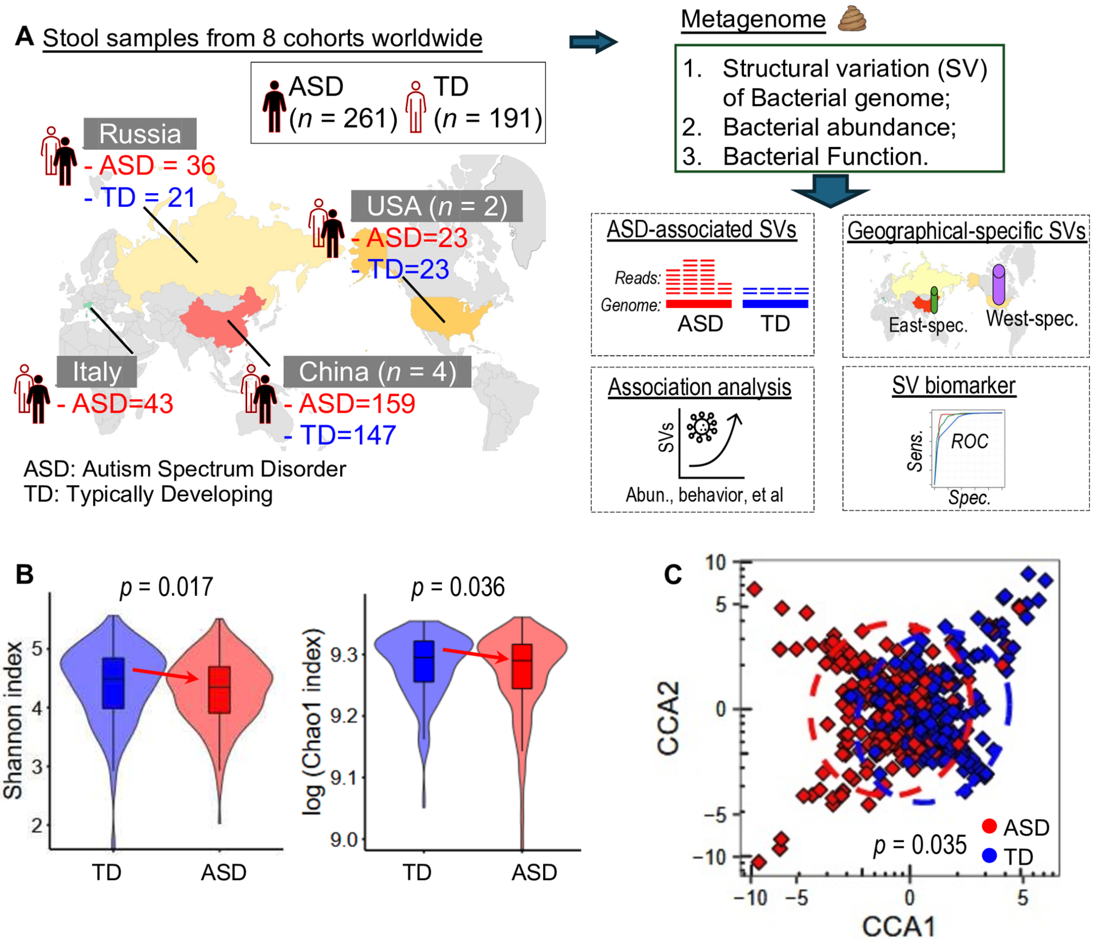

Figure 1 Study design and analysis workflow. (A) Stool samples collected from children with autism spectrum disorder (ASD) and children with typically developing (TD) status. Summary of the workflow including analysis of ASD-associated bacterial structural variants (SVs), identification of geography-specific SVs, association analysis and identification of ASD diagnostic panel. (B) Overall bacterial diversity (Shannon and Chao1 index) between ASD and TD. Significance was tested by a two-tailed Wilcoxon Rank Sum test. (C) Overall bacterial composition profile between ASD and TD (based on Horn-Morisita index). Significance was tested by permutational multivariate analysis of variance (PERMANOVA). ASD, Autism Spectrum Disorder; CCA, constrained correspondence analysis; d-SV, deletion SV; SV, structural variation; TD, typically developing; v-SV, variable SV.

health and supporting proper brain function[18]), and carbohydrate metabolism, as well as critical cellular processes for regulating bacterial growth such as gene transcription and translation. Hence, our findings suggest that the functional consequences of bacterial genomic SVs may be involved in the pathogenesis of ASD.

The functional outcomes of bacterial SVs align with gut metabolic dysregulation in ASD

We subsequently used the Unified Metabolic Analysis Network (*HUMAnN*) pipeline to assess the functional potential and activity of gut bacteria in ASD and neurotypical children. Compared with neurotypical children, those with ASD exhibited multiple dysregulated metabolic pathways (online supplemental figure S2D), including iron homeostasis (eg, ASD-depleted ferritin, a critical component of numerous enzymes involved in neurotransmitter synthesis[19]), amino acid metabolism (eg, an elevated glutamate-gamma-aminobutyric acid antiporter system, previously linked to social and sensory impairments in ASD[20]), as well as signalling and cellular processes. These metabolic disruptions were strongly associated with SVs in gut bacteria, particularly within *Ba. uniformis* and *R. torques* (figure 2C, D). Together, these findings suggest a potential link between gut bacterial genomic alterations and the metabolic dysregulation implicated in ASD pathophysiology.

Ba. uniformis genomic SVs encompass genes involved in thiamine and ferritin metabolic regulation

Given the potential functional significance of *Ba. uniformis* in ASD as shown above, we conducted an indepth investigation into this bacterium and focused on its genomic architecture. A total of 146 SVs, comprising 20 v-SVs and 126 d-SVs, were identified in *Ba. uniformis* genome. Correspondence analysis revealed that the SV profile of *Ba. uniformis* in ASD was significantly different from that in neurotypical children ($p<0.05$) (figure 3A). In particular, a v-SV spanning the 1689–1708 kb genomic region of *Ba. uniformis*, which encompasses genes associated with ferritin and thiamine regulation, was significantly less in ASD than in neurotypical children ($p=0.036$) (figure 3B). Concurrently, one of the d-SVs located at the 1–3 kb region of *Ba. uniformis*, containing genes associated with exopolysaccharide (EPS) biosynthesis, was depleted in ASD compared with neurotypical children (ORs=0.55, $p=0.01$; Fisher exact test) (online supplemental figure S3B).

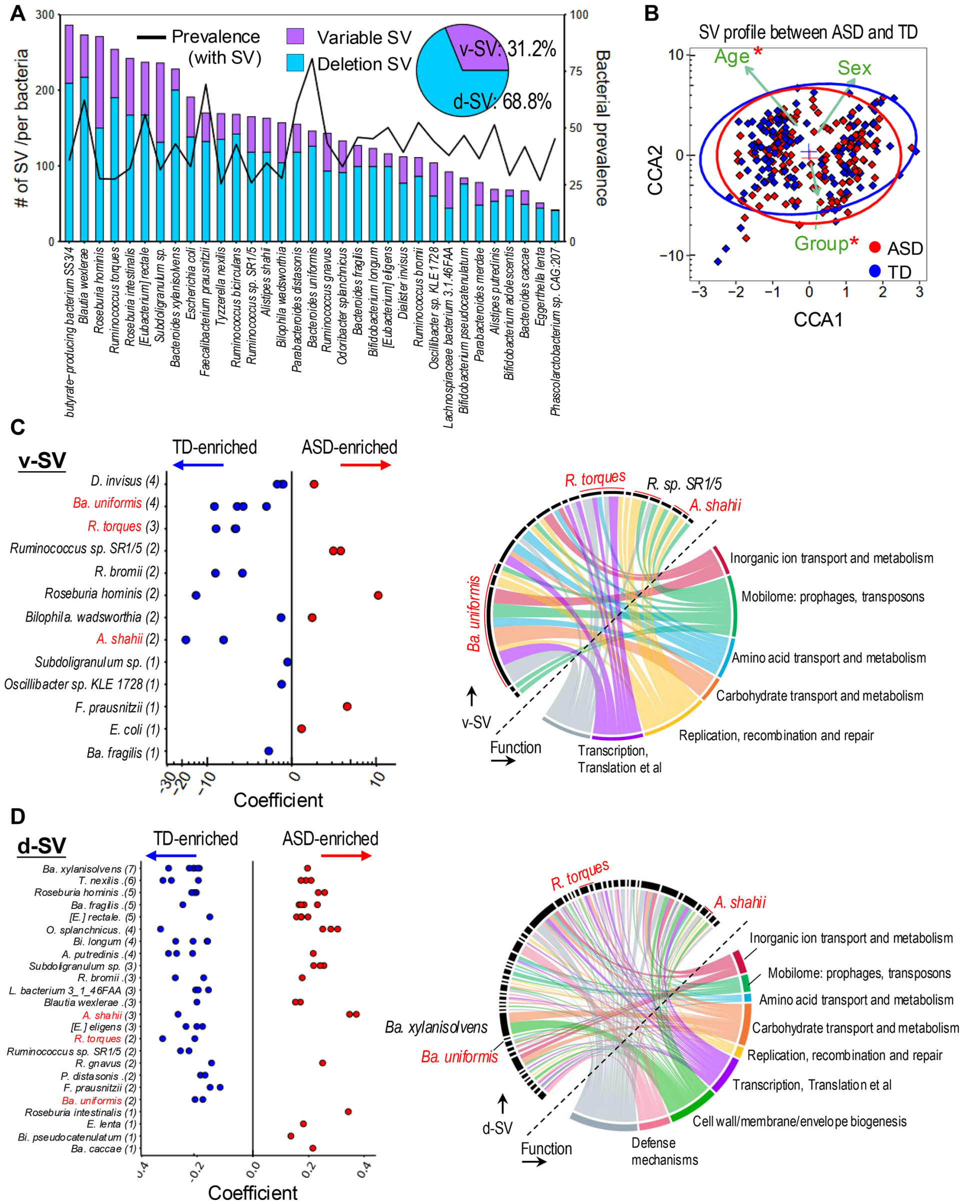

Figure 2 Analysis of ASD-associated bacterial SVs. (A) The distribution of bacterial genomic SVs in each bacterium in the children's gut. Two types of SVs were analysed: variable SV (v-SV) and deletion SV (d-SV). (B) Overall bacterial SV profiles between ASD and TD. Significance was tested by ANOVA-like permutation test. *p<0.05. (C) Panel left: the differential genomic v-SVs between ASD and TD were detected by a linear mixed-effects model adjusted for age, sex and cohort effects. Panel right: the functional annotation of the v-SVs. (D) Panel left: the differential genomic d-SVs between ASD and TD. Panel right: the functional annotation of the differential d-SVs. In (C–D) Red circle point represents ASD-enriched SV, while blue circle point represents ASD-depleted SV. Coefficient was calculated by a linear mixed effect model adjusted for cohort effects, age and sex. ANOVA, analysis of variance; ASD, autism spectrum disorder; CCA, constrained correspondence analysis; Cor, Spearman correlation; pCor, partial Spearman correlation; SV, structural variation; TD, typically developing.

These findings suggest that genomic alterations in *B. uniformis* may affect several functional processes, such as nutrient metabolism and iron homeostasis, potentially influencing ASD development.

Validation of *B. uniformis* genomic SVs in association with ASD-associated behaviour in a humanised microbiome mouse model

To validate our in-silico findings, we retrieved publicly available dataset[8] generated from a humanised microbiome model, where germ-free mice were colonised with faecal microbiota obtained from children with ASD or neurotypical children (online supplemental figure S3C). We analysed stool samples and behavioural data collected from the weaned offspring of these humanised mice, referred to as oASD mice (offspring of mice colonised with ASD microbiota) and oTD mice (offspring of mice colonised with TD microbiota). Compared with oTD mice, we observed a depleted level of the v-SV spanning the 1689–1708 kb genomic region of *Ba. uniformis* in oASD mice (online supplemental figure S3D). Notably, the level of this v-SV was positively correlated with the social interaction in mice, while its depletion was negatively correlated with mouse repetitive and persistent behaviours (figure 3C). These findings thus experimentally validated our computational predictions, demonstrating that ferritin-related and thiamine-related genomic alterations in

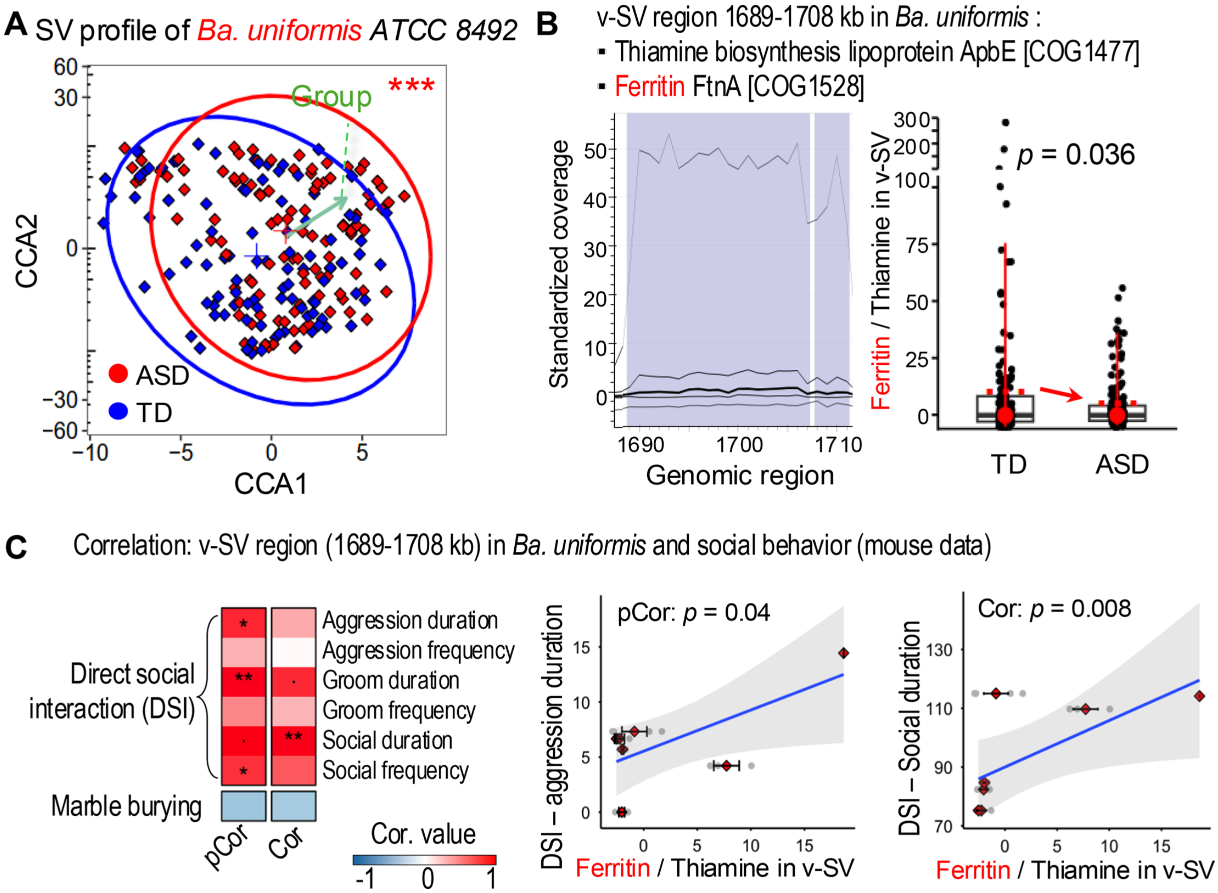

Figure 3 ASD-associated SVs in *Bacteroides uniformis*. (A) Overall SV profiles within *B. uniformis* between ASD and TD. Significance was tested by ANOVA like permutation test. *** p<0.001. (B) The v-SV spanning the 1689–1708 kb genomic region of *B. uniformis* (panel left), and its level between ASD and TD (panel right). (C) The correlation between the v-SV (1689–1708 kb region) of *B. uniformis* and mouse behavioural phenotypes (eg, DSI and marble burying (ie, repetitive/persistent behaviours)). ANOVA, analysis of variance; ASD, autism spectrum disorder; CCA, constrained correspondence analysis; Cor, Spearman correlation; pCor, partial Spearman correlation; SV, structural variation; TD, typically developing; v-SV, variable SV.

Ba. uniformis are functionally linked to host behavioural phenotypes, thereby bridging microbial genomic variation with ASD-associated neurobehavioural outcomes.

Toxin-antitoxin system-associated SVs are associated with the enrichment of the pathobiont *R. torques* in ASD

R. torques is an ASD-related pathobiont,[17] whose abundance was enriched in ASD compared with neurotypical children (figure 4A, online supplemental figures S1F, S3E). The overgrowth of *R. torques* in ASD remained unclear. Here, we examined the genomic architecture of *R. torques* and identified 254 SVs in the *R. torques* genome, comprising 64 v-SVs and 190 d-SVs. Correspondence analysis revealed significant differences in the SV profiles between ASD and neurotypical children ($p<0.05$) (figure 4B). Particularly, we identified a v-SV within the 1328–1331 kb region of *R. torques* that encodes the growth inhibitor *MazF*, which was significantly depleted in ASD compared with neurotypical children ($p=0.036$) (figure 4C). *MazF* is a key toxin component of the *MazF-MazE* toxin-antitoxin system and exhibits ribonuclease activity critical for bacterial growth regulation.[21] Intriguingly, we found that the level of this *MazF*-containing v-SV was negatively correlated with the abundance of *R. torques* (figure 4D). Meanwhile, the abundance of the antitoxin *MazE* showed no significant difference between ASD and neurotypical children (online supplemental figure S3F). These correlative findings suggest that the depletion of *MazF*-containing v-SV in *R. torques* is associated with higher abundance of *R. torques* in ASD, potentially through reduced growth inhibition. This at least provides a plausible regulatory mechanism linking genomic variation to bacterial proliferation in the context of ASD.

ASD-associated microbial genomic SVs are geographically different

The inclusion of geographically diverse populations in this dataset enabled a systematic investigation of geography-associated SVs in ASD children. Using a linear mixed-effects model adjusted for age, sex and cohort effects, we identified 38 v-SVs and 100 d-SVs that distinguished ASD from neurotypical children in Western populations, while 41 v-SVs and 85 d-SVs were found in Eastern populations (figure 5A, B and online supplemental tables S5,6). Cross-population analysis revealed 13 v-SVs and 45 d-SVs shared between both groups (figure 5C and online supplemental figure S4A). For instance, an ASD-depleted v-SV in *Ba. xylanisolvens* (5027–5046 kb) encoded the antirestriction protein that suppresses bacterial antiphage defenses[22] (figure 5D and online supplemental figure S4B). We also identified an ASD-enriched d-SV in *Eubacterium rectale* (1299–1301 kb), harbouring nitroreductase genes with potential applications in prodrug metabolism for anticancer therapy[23] (figure 5E and online supplemental figure S4C). Population-specific signatures featured 25 v-SVs and 55 d-SVs unique to

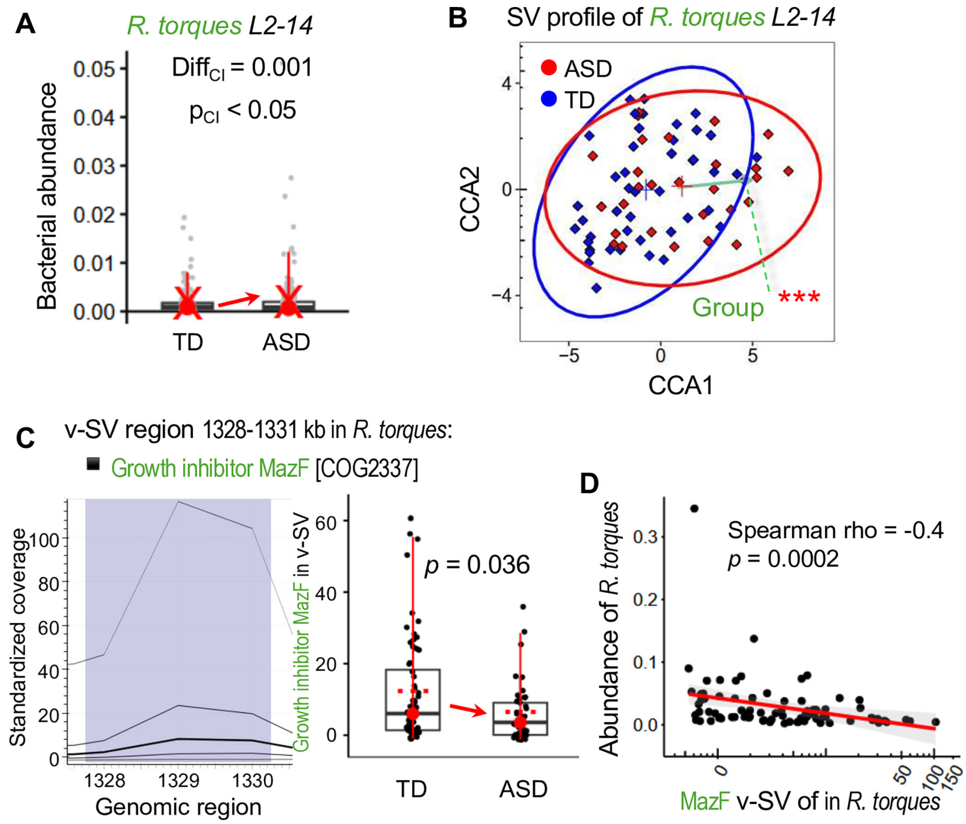

Figure 4 ASD-associated SVs in *Ruminococcus torques*. (A) The abundance of *R. torques* between ASD and TD. Significance was tested by a differential Core Index analysis (online supplemental figure S1). (B) Overall SV profiles within *R. torques* between ASD and TD. Significance was tested by ANOVA-like permutation test. ***p<0.001. (C) The v-SV spanning the 1328–1331 kb genomic region of *R. torques* (panel left), and its level between ASD and TD (panel right). (D) The Spearman correlation analysis between *R. torques* genomic variant (1328–1331 kb region) and the bacterial abundance. ANOVA, analysis of variance; ASD, autism spectrum disorder; CCA, constrained correspondence analysis; CI_{diff}, Core Incore index difference; SV, structural variation; TD, typically developing; v-SV, variable SV.

Western children, such as a d-SV in *Bi. adolescentis* (128–133 kb) encoding the fimbrial subunit *FimA* (figure 5F), a critical adhesin for mucosal colonisation.[24] Conversely, Eastern children exhibited 28 v-SVs and 41 d-SVs, exemplified by a d-SV in *Alistipes putredinis* (630–632 kb) encompassing genes for the *AraC* transcriptional regulator (figure 5G), a key mediator of carbohydrate metabolism and bacterial stress adaptation.[25] Together, these findings reveal conserved patterns of microbial genomic disruptions in ASD while highlighting region-specific genomic adaptation, suggesting that geography-associated SVs might play an important role in ASD pathophysiology.

SV biomarkers distinguish ASD children from neurotypical children

While bacterial markers have been previously explored for ASD diagnosis,[6] we sought to investigate the diagnostic potential of bacterial genomic SVs as novel biomarkers. To this end, we implemented a rigorous analytical framework using our multicohort dataset (figure 6A). We established a discovery cohort comprising 70% of samples (n=338) randomly selected from all eight cohorts, while the remaining 30% of samples (n=79) were reserved as an independent validation cohort. We first trained the biomarker panel for ASD diagnosis with pure v-SVs (n=26 differential v-SVs with age and sex adjusted; figure 2C), pure d-SVs (n=74; figure 2D) and pure bacteria (n=93; online supplemental figure S1F). Using a sequential floating forward selection (SFFS) strategy combined with support vector machine (SVM) classification, we identified optimal biomarker panels consisting of 11 v-SVs, five d-SVs and nine bacterial species that demonstrated the best diagnostic performance, respectively (online supplemental figure S5A–C). These panels achieved area under the receiver operating characteristic curve (AUROC) values of 79.1% (v-SVs), 75.2% (d-SVs) and 72.3% (bacteria) in distinguishing ASD children from neurotypical children, respectively. Compared with panels using pure SVs or bacteria alone, the mixture integrating both SVs (six v-SVs and three d-SVs; eg, 1328–1331 kb region of *R. torques*) and bacterial species (n=3; eg, *C. bolteae*) (figure 6B and online supplemental figure S5D), the 12-marker panel demonstrated significantly improved diagnostic performance, with an AUROC of 85.8% in the discovery cohort (figure 6C) and an AUROC of 81.1% in the independent validation cohort (figure 6D). To further evaluate the generalisability of this 12-marker panel, we implemented a cross-by-cohort validation strategy (figure 6A), which demonstrated consistent diagnostic performance, with a median AUROC of 72.0% (Q1=67.3%, Q3=73.0%) (figure 6E), supporting the robustness of combined biomarker panel across diverse cohorts.

To account for potential population-specific variations, we developed two distinct ASD diagnostic models tailored for Western and Eastern children. For the Western cohort, a panel comprising 10 markers (seven SVs and three bacterial species) achieved an AUROC of 79.5% in distinguishing ASD children from neurotypical controls (online supplemental figure S5E). Similarly, for the Eastern cohort, a panel of nine markers (seven SVs and two bacterial species) showed stronger performance with an AUROC of 81.5% (online supplemental figure

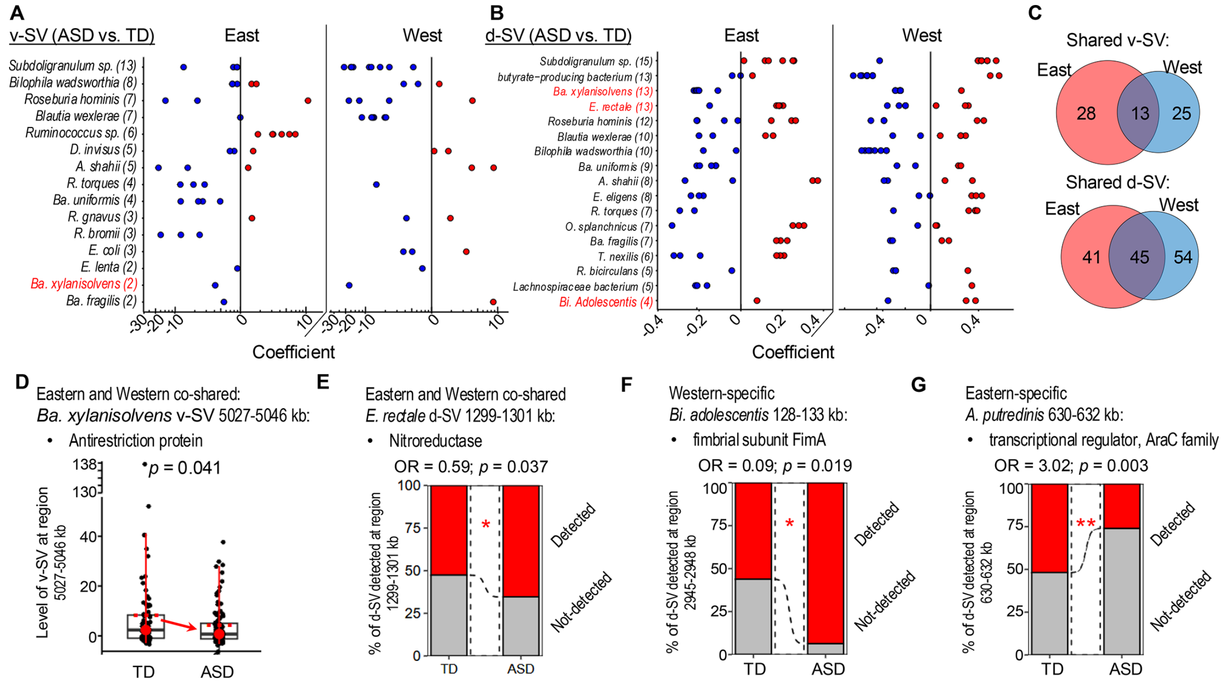

Figure 5 ASD-associated microbial genomic SVs are geographically different. (A) v-SVs that are differentially distributed between ASD and TD in the Eastern (panel left) and Western children (panel right). (B) d-SVs that are differentially distributed between ASD and TD in the Eastern (panel left) and Western children (panel right). In (A–B), the red circle point represents ASD-enriched SV, while the blue circle point represents ASD-depleted SV. The coefficient was calculated by a linear mixed effect model. (C) The geography-shared and region-specific v-SVs or d-SV. (D) The *Bacteroides xylanisolvens* (*Ba. xylanisolvens*) genomic variant (v-SV: 3541–3556 kb) in ASD and TD. (E) The *Eubacterium rectale* (*E. rectale*) genomic variant (d-SV: 1299–1301 kb) in ASD and TD. (F) The *Bifidobacterium adolescentis* (*Bi. adolescentis*) genomic variant (d-SV: 128–133 kb) in ASD and TD. (G) The *Alistipes putredinis* (*A. putredinis*) genomic variant (d-SV: 630–632 kb) in ASD and TD. The significance of d-SV between ASD and TD was tested by Fisher exact test. ASD, autism spectrum disorder; d-SV, deletion SV; SV, structural variation; TD, typically developing; v-SV, variable SV.

S5F). These results established bacterial genomic SVs as valuable biomarkers for ASD diagnosis and highlighted the superior diagnostic accuracy achieved by combining SV markers with bacterial abundance data, compared with traditional approaches relying exclusively on either SVs or bacterial species.

DISCUSSION

In this study, we performed the first meta-analysis to systematically investigate the association between bacterial genomic variation and ASD. Using a multicohort approach, we analysed stool samples from 452 children worldwide and identified significant bacterial genomic SVs linked to metabolic dysregulation in ASD compared with neurotypical children. These findings were further supported by the functional roles of genes within the identified bacterial SVs, thus providing new insights into how bacterial genomic variations may influence metabolic regulation in ASD. Given that bacterial genomic variations can profoundly alter bacterial gene function, metabolic pathways and host-microbe interactions,[26] they may potentially result in increased gut permeability and disruption of gut-brain axis, thereby contributing to ASD pathogenesis.

The identified bacterial SVs associated with ASD and their functional consequences illustrated the utility of SV-based association analysis in uncovering host-microbe interactions at both functional and mechanistic levels. Our functional predictions revealed multiple dysregulated metabolic pathways in ASD compared with TD (online supplemental figure S2D). This is consistent with recent findings from ASD mouse models showing that systemic host metabolic disruption including lipid metabolism is a pathological feature in ASD.[4 26 27] Moreover, our analysis linked ASD-associated SVs to genes involved in key metabolic processes (eg, amino acid metabolism) (figure 2C, D). This connection provides a potential mechanistic link, suggesting that bacterial genomic variations may contribute to the observed gut and systemic metabolic dysregulation in ASD. For instance, we identified multiple genomic SVs within *Ba. uniformis*, including a v-SV implicated in thiamine and iron regulation (1689–1708 kb region) and a d-SV involved in EPS biosynthesis (1–3 kb region). Given the essential role of thiamine and iron metabolism in neurological function,[14] dysregulation of this pathway may contribute to the metabolic and neurological disturbances observed in ASD. Experiments in mice supported the functional relevance of these SVs, demonstrating a strong association between the thiamine/iron-related SV and behavioural phenotypes, including altered social interaction frequency and increased repetitive behaviours. EPS biosynthesis plays a critical role in bacterial adherence, cohesion and accumulation on the gut mucosa, facilitating biofilm formation and microbial colonisation.[28] *Ba. uniformis* is generally regarded as a beneficial commensal bacterium, which contributes to gut homeostasis and host health. However, dysfunction in EPS biosynthesis in *Ba. uniformis* may disrupt its interaction with the host, potentially diminishing its beneficial effects.[29] Collectively, these SVs in *Ba. uniformis* may correlate with metabolic dysregulation and perturb microbial ecology, ultimately undermining its protective role in gut health.

Similarly, we identified genomic alteration in the *R. torques* genome, including a v-SV encoding growth inhibitor *MazF* (1328–1331 kb region), the toxin element of the *MazF-MazE* toxin-antitoxin system.[21 30 31] The depletion of this v-SV may

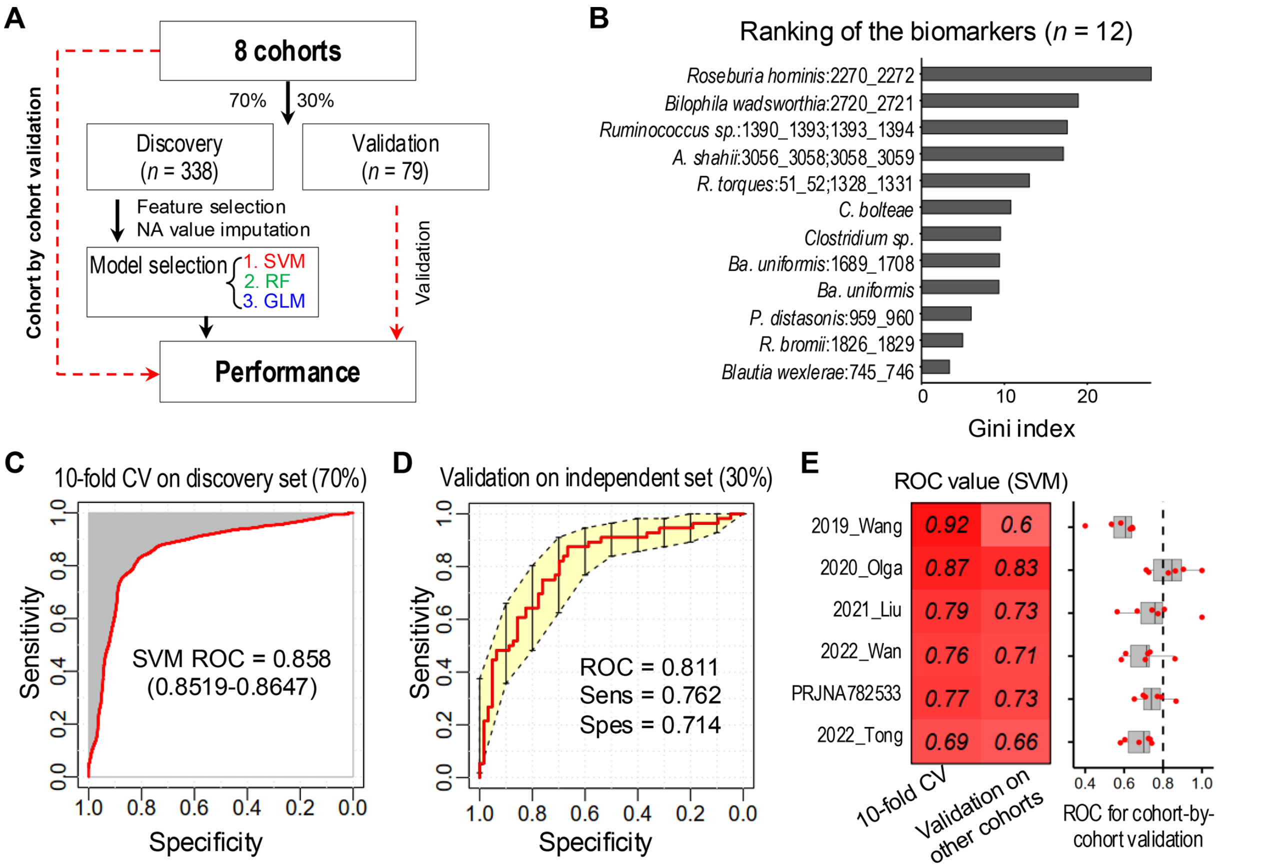

Figure 6 Identification of ASD diagnostic panel. (A) The analytical framework for the identification of ASD diagnostic panel. (B) The ranking of the identified bacterial markers and bacterial genomic SV markers (ie, 12-marker panel). The impotence of each maker was measured by Gini index. (C) The performance (ie, area under the receiver-operating characteristic curve (ROC)) of the 12-marker panel trained by support vector machine (SVM) algorithm in the discovery cohort. The 10-fold cross-validation strategy was applied. (D) The performance of the 12-marker panel in the independent validation cohort. (E) A cross-by-cohort validation strategy was applied to validate the generalisability of this 12-marker panel. ASD, autism spectrum disorder; GLM, generalised linear model; RF, random forest; SV, structural variation.

reduce growth inhibition, potentially leading to bacterial overgrowth. This aligns with the observed enrichment of *R. torques* in ASD compared with neurotypical children, as reported in both previous studies[17] and our current analysis (figure 4A and online supplemental figure S1F), as well as in two independent mouse models (online supplemental figure S3E). While we identify a compelling genomic association, future functional studies are required to establish a direct causal role for the *MazF-MazE* system in modulating *R. torques* abundance in the gut ecosystem. Meanwhile, we identified a d-SV within *R. torques* (266–271 kb region) that encompasses genes encoding α-galactosidase (online supplemental figure S3G). α-galactosidase is a key glycosyl hydrolase enzyme involved in the degradation of mucin glycans.[32] Such genomic alteration in *R. torques* suggests potential impairments in mucin glycan processing, which could compromise gut barrier integrity and disrupt the gut-brain axis. Here, the depletion of growth-inhibiting SVs (eg, *MazF*-encoding v-SV) could promote bacterial overgrowth, while the dysfunction of mucin-degrading enzymes (eg, α-galactosidase-encoding d-SV) may impair gut barrier function and microbial homeostasis. These genomic alterations likely disrupt the gut ecosystem, potentially exacerbating gut-brain axis dysfunction and contributing to ASD. Taken together, the findings highlight the importance of bacterial genomic variation in understanding the complex interplay between gut microbiota and neurodevelopmental disorders.

We next studied the critical role of geographic and demographic factors in shaping the gut microbiome and their potential contribution to ASD. A key limitation is the current scarcity of shotgun metagenomic data from ASD cohorts outside of China within Eastern populations, as most available studies mainly rely on 16S ribosomal RNA sequencing. Nevertheless, the population-specific signatures we observed are consistent with well-established evidence that regional variations in diet, lifestyle and environmental exposures exert selective pressures on gut bacteria.[33 34] This can drive microbial genomic diversification, which may enhance adaptability while potentially entrenching dysbiotic states that exacerbate ASD phenotypic traits. Our analysis revealed both conserved and region-specific bacterial genomic variants associated with ASD. Notably, we identified a conserved ASD-depleted variant (5027–5046 kb) in *Ba. xylanisolvens* (a probiotic with documented anticolitis properties[35]) encoding a phage defence-related antirestriction protein. This depleted genomic variant could facilitate lytic infection in bacterial cells, reducing bacterial abundance in ASD (online supplemental figure S1F), thereby potentially diminishing the bacterium's ability to exert its protective effects. In Western populations, we identified a d-SV in *Bi. adolescentis* (128–133 kb) encoding mucosal colonisation-enhancing *FimA* adhesin, likely reflecting Western dietary adaptations, as increased mucosal adhesion may improve bacterial retention in environments with limited dietary fibre.[36] Meanwhile, in

Eastern populations, an *AraC*-regulator d-SV in *A. putredinis* (630–632 kb) was associated with carbohydrate metabolism and stress response, potentially indicative of an adaptation to traditional diets.[37] These findings demonstrate how regional selective pressures drive microbial genomic diversification, highlighting the need to consider geographic factors in ASD research and the potential for developing region-specific diagnostic and therapeutic approaches.

Beyond mechanistic insights, our study demonstrated the diagnostic potential of bacterial genomic biomarkers for ASD. Compared with species-abundance metrics that provide taxonomically coarse signal, the genomic SVs offer superior biological precision by distinguishing functionally divergent strains within a species, yielding more interpretable and potentially causal biomarkers for early ASD detection. This specificity not only enhances biological insight but also paves the way for simpler, targeted clinical assays, thereby establishing a direct translational path. To this end, we developed a diagnostic panel integrating bacterial SVs and bacterial abundance data, achieving an AUROC of 85.8% and 81.1% in the discovery and validation cohort, respectively, and outperforming models based solely on bacterial abundance. A previous meta-analysis showed that while leading blood-based proteomic and metabolomic approaches report pooled/predictive AUCs of approximately 88%, individual study results vary widely (AUC range: 57.0–99.9%).[38] Our robust, stool-based method offers the practical advantage of being non-invasive, which is particularly beneficial for paediatric populations, where repeated blood draws are challenging, and it enables feasible longitudinal sampling to study microbiome dynamics and intervention effects. We further identified population-specific diagnostic panels for Western and Eastern cohorts, with AUROCs of 79.5% and 81.5%, respectively. While our findings are relatively robust within the studied population, the inherent clinical and biological heterogeneity of ASD, which likely contributed to performance variability, underscores the necessity for future validation across broader age ranges, sex ratios and phenotypic subgroups to establish the generalisability of these biomarkers across the full autism spectrum.

While our findings provided new evidence for the role of bacterial genomic variation in ASD, several limitations should be acknowledged. First, the cross-sectional nature of our study limits the ability to infer causal relationships between bacterial SVs and ASD, and future longitudinal studies are needed to determine whether these genomic variations precede or result from ASD-related physiological changes. Second, our analysis focused solely on the bacterial metagenome and did not include host whole-exome or whole-genome sequencing, preventing assessment of host genetic contributions, a major component of ASD aetiology and their potential interactions with the gut microbiome. Future studies integrating host genomics and microbial genomics will be essential to elucidate these host-microbe interactions more comprehensively. Finally, the diagnostic panels developed in this study require further validation in larger, more diverse cohorts to ensure generalisability across different populations and age groups.

In conclusion, this study provided the first comprehensive evidence linking bacterial genomic SV to ASD in children, highlighting a potential role of gut microbial genomic alterations in neurodevelopmental disorder. Moreover, we developed a diagnostic approach for ASD by combining microbial genomic and taxonomic profiling, which may enable more precise discovery and clinical stratification for children with ASD.

METHODS

Study subjects

A total of 452 children comprising 261 ASD and 191 neurotypical children from an in-house cohort (n=128) and published faecal metagenomic datasets of 4 different countries/cities (n=324) were included in this study. Faecal samples collected from the in-house cohort consist of 64 ASD children recruited from a regional teaching hospital (Prince of Wales Hospital, Shatin, Hong Kong) and 64 neurotypical children with matched age and sex from the same catchment area within the same period. Children with ASD were diagnosed by paediatricians or clinical psychologists according to the standard of the Diagnostic and Statistical Manual of Mental Disorders. Exclusion criteria included: chronic seizures; recent infections (within 1 month); conditions affecting diet or physical activity; use of antibiotics, antifungals, prebiotics or probiotics within 1 month prior to the study; participation in clinical trials or dietary interventions within the past month. The study protocol was performed in compliance with the Declaration of Helsinki and approved by the Clinical Research Ethics Committee of The Chinese University of Hong Kong (CREC Ref. No.: 2014.026 and 2016.607).

We also searched for ASD studies published up to 2024 on PubMed and identified seven studies that have used shotgun metagenomic sequencing on children's faecal samples. Metagenomic datasets were downloaded from the following sources: 2019_Gil,[8] 2019_Wang,[39] 2020_Olga,[40] 2021_Liu,[41] PRJNA782533, 2022_Tong[42] and 2024_Luigi.[43] Detailed sample sizes and demographic characteristics for each cohort are summarised in online supplemental table S1. Clinical information, including diagnostic criteria and metadata such as age, sex and geographic characteristics, was extracted from the supplementary tables of the original articles. These studies represent a diverse range of geographic regions and populations (Western: USA, Italy and Russia; Eastern: China), allowing us to explore potential geographic influences on the gut microbiome in ASD. Meanwhile, our in-house cohort was constrained in age (median (Q1,Q2) =5[5 6] years) and comprised a high male-to-female ratio (4.82:1), aligning with the demographic profiles of major publicly available datasets (online supplemental table S1), thereby reducing variance from these key confounds for biomarker identification. All studies included had previously obtained approval from their respective ethics institutional review boards.

To further validate our findings, we also incorporated data from two mouse studies: 2019_Gil[8] and 2022_You.[44] To maximise comparability and reproducibility, we reprocessed all sequence data using a standardised bioinformatics pipeline including quality control, read trimming, taxonomic and functional profiling and identification of SVs, ensuring consistency in data processing and analysis across studies.

Shotgun metagenomic sequencing

Total DNA was extracted from faecal samples by Maxwell RSC PureFood GMO and Authentication Kit (Promega) according to the manufacturer's instructions. Shotgun metagenomic sequencing was performed at Novogene (Beijing, China) by Illumina Novoseq 6000 platform with paired-end 150 bp.

Profiling of microbial abundance and pathways

For quality control of all metagenomic sequencing data, we employed *KneadData* (V.0.7.2) with default parameters. This tool is specifically designed to filter out contaminated reads originating from the host genome or other non-microbial sources, ensuring that only high-quality microbial reads are retained for downstream analysis. By removing extraneous sequences, *KneadData* enhances the accuracy and reliability of subsequent taxonomic and functional profiling. To infer bacterial abundance, we used *MetaPhlAn* (V.4.0.2) with default parameters[45] for species-level microbial profiling. To assess the functional potential and activity of gut bacteria, we used the *HUMAnN* pipeline with default parameters to reveal the microbial pathways.

Community analysis, including alpha diversity and beta diversity, was calculated with the phyloseq R package. Alpha diversity was evaluated by Shannon and Chao1 index. Beta diversity was measured by Horn-Morisita distance, and the constrained correspondence analysis (CCA) was used for ordination analysis. Community dissimilarities were tested by PERMANOVA with 1000 iterations using the Horn-Morisita distance.

Bacterial genomic SVs and the function annotation

We applied *SGV-Finder* (V.0.1.0), a specialised tool designed to detect and profile SVs in metagenomic data for the identification of bacterial genomic SVs.[46] *SGV-Finder* identifies two main types of SVs: d-SVs and v-SVs, which represent genomic regions with significant variations in coverage or sequence composition. To achieve this, we first performed the iterative coverage-based read assignment algorithm (v1.366), which aligns sequencing reads to bacterial reference genomes obtained from the *proGenomes* database (http://progenomes1.embl.de/). The core algorithm of SGV-Finder was then executed to analyse coverage patterns across the microbial genomes. Specifically, the tool divided each bacterial genome into 1 kb bins and evaluated the coverage within these regions to identify statistically significant variations. Regions with abnormal coverage patterns, such as deletions or variable regions, were flagged as potential SVs.

To identify d-SVs, we analysed metagenomic bin coverage across samples. Regions absent in some individuals but retained in others were detected by generating coverage histograms per microbe and sample. A trough distinguishing near-zero coverage bins from those with median coverage (≥10 reads) was identified. Bins with coverage below this trough were considered deleted, whereas those above were retained. d-SVs were defined as bins deleted in 25–75% of samples.

To detect v-SVs, bins deleted in >95% of subjects were excluded for each microbial species. Coverage variation in remaining bins was assessed by standardising coverage per sample (mean subtraction, SD division). A beta-prime distribution was fitted for each bin, and bins with coverage values in the top fifth percentile were classified as v-SVs.

Representative genome sets (FASTA files containing bacterial gene sequences) were downloaded from the *proGenomes* database. For each SV of interest, we retrieved the corresponding gene/protein IDs. Functional annotation of these IDs was performed using eggNOG (evolutionary gene genealogy Non-supervised Orthologous Groups)-mapper,[47] which assigns protein-coding sequences to precomputed orthologous groups within the eggNOG database. To obtain detailed functional descriptions, the annotated gene/protein IDs were subsequently queried against the National Center for Biotechnology Information (NCBI) databases using the *entrez_link* function from the R package *rentrez* (V.1.2.3). Final gene annotations, combining the product description from NCBI and the functional category Clusters of Orthologous Genes (COG) from eggNOG, were then manually summarised and curated (online supplemental tables S3b, S4b). The Circos plot was generated using the R package *circlize* (V.0.4.14).

Beta diversity was measured by Horn-Morisita and Jaccard distance for v-SVs and d-SV profiles, respectively. The CCA was used for ordination analysis. Community dissimilarities were tested by permutation test for CCA with 1000 iterations.

Differential core bacteria identification based on multicohort analysis

We aimed to identify the differential core bacteria between ASD and neurotypical children. To achieve this, we developed a Core Index (CI) considering the bacterial prevalence, abundance and consistency among different cohorts. The proposed CI of the i-th bacteria was calculated as follows:

$$CI_i = w_i \times RA_i \times F_i$$

CI difference (CI_{diff}) for the i-th bacteria between ASD and neurotypical children was defined as:

$$CI_{diff} = CI_{CRC} - CI_{CTRL}$$

where RA_i and F_i are the relative abundance and prevalence of the i-th bacteria, respectively. w_i is a factor considering the existence of i-th bacteria in different cohorts. Bacteria with a prevalence >25% in an individual cohort or environment source were considered as present on that cohort.

To identify the significance of CI_{diff}, we obtained the overall CI_{diff} density and fitted appropriate mathematical distributions on the densities according to the observed density shape. A p value was calculated by the cumulative probability $p\left(CI \geq q\right)$. The significance level α was set to 0.05.

Differential SVs between ASD and neurotypical children

To identify bacterial SVs that are differentially distributed between ASD and neurotypical children, we employed a linear mixed-effects model. This statistical approach is particularly well-suited for multicohort analyses, as it accounts for potential confounding factors, including cohort effects, age and sex. The model was designed to assess the association between disease status (ASD vs neurotypical children) and bacterial SVs using the following formula

$$SV \sim disease\ status + Age + Sex + Condition\left(cohort\right)$$

where, Condition(cohort) accounts for variability across different cohorts, ensuring that the results are not biased by study-specific effects.

To identify differential SVs, we applied stringent statistical criteria: (1) the p value for the coefficient of disease status had to be <0.05, indicating a significant association between the SV and ASD; (2) the p values for the coefficients of age and sex had to be >0.05, ensuring that the identified SVs were independent of these demographic factors. P values were adjusted using the Benjamini-Hochberg false discovery rate (FDR) correction. SVs meeting these criteria were considered differentially distributed between ASD and neurotypical children. This approach allowed us to control for cohort heterogeneity and demographic variables, ensuring robust and reliable identification of ASD-associated SVs.

Mouse colonisation and behaviour testing

To validate our in-silico findings linking *Ba. uniformis* genomic SVs to ASD status and associated behaviours, we used a mouse model from a previous study (online supplemental figure S3C).[8] Briefly, 3–4-week-old germ-free C57BL/6J mice were colonised with faecal samples from human donors (children with ASD and neurotypical children). Mice (4–6 females and 2–3 males per donor) were stool-gavaged and later mated (3 weeks) according to their respective human donor groups. Pregnant dams were single-housed, and offspring born within a 1-week window were grouped by sex (4–5 mice per cage) for subsequent analyses. All animal experiments were approved by the Animal Experimentation Ethics Committee of The Chinese University of Hong Kong.

Behavioural tests, including the marble burying (MB) and direct social interaction (DSI) tests, were initiated at 6 weeks of age. Mice were acclimated overnight to the testing room and allowed to rest for at least two nights after cage changes. All behavioural assessments were conducted during the light phase, with tests spaced 4–7 days apart. To minimise time-of-day effects, testing order was alternated across experimental groups.

The MB test was conducted in a standard cage. Mice were habituated to the cage for 10 min, temporarily removed and then returned to the same cage after levelling the bedding and placing 20 glass marbles (arranged 4×5) on the surface. After 10 min, the number of marbles buried was recorded and photographed.

The DSI test was employed to assess sociability. Each test mouse was placed in a clean, empty cage for a 10 min habituation phase, with grooming behaviour scored during the final 5 min. A stimulus mouse was then introduced for 6 min. Interactions were recorded overhead and scored using ETHOM software for social approach, aggression and grooming.

ASD diagnostic panels consisting of both bacterial SVs and bacterial species

To develop an ASD diagnostic panel, we employed machine learning approaches based on species-level bacterial abundance and bacterial SVs. Our analysis began by establishing a discovery cohort comprising 70% of the samples (n=338), which were randomly selected from all eight cohorts. The remaining 30% of samples (n=79) were reserved as an independent validation cohort, ensuring an unbiased evaluation of the diagnostic model's performance. For feature selection, we used the SFFS strategy, a robust method that iteratively adds and removes features to identify the optimal subset of bacterial species or SVs with the highest predictive power. To identify the best-performing model, we evaluated three machine learning algorithms including SVM, random forest and generalised linear model by the *caret* R package. Each algorithm was combined with a 10-fold cross-validation strategy to ensure robust performance estimation and avoid overfitting. During cross-validation, the dataset was partitioned into 10 subsets, with the model trained on nine subsets and validated on the remaining subset. This process was repeated 10 times, with each subset used once as the validation set, ensuring that the performance metrics were averaged across all folds for reliability. The final step involved selecting the optimal machine learning algorithm based on its performance metrics (ie, AUROC) using the top features identified by the SFFS strategy. The SVM emerged as the most effective algorithm for our dataset. This final model was trained on the discovery cohort and subsequently validated on the independent validation cohort.

To evaluate model generalisability and diagnostic accuracy, a cross-by-cohort validation strategy was employed. This approach used a study cohort (in-house and external ASD studies; n=6 cohorts in total) to train an initial model based on bacterial abundance and SVs (n=12). The model was then tested, without retraining or re-optimisation, on the remaining independent study cohorts. Performance was assessed using AUROC.

Statistical analysis

Data were presented as mean±SD for normally distributed variables or median (first quartile, third quartile) for non-normally distributed variables, as appropriate. Statistical comparisons were performed using a two-tailed Mann-Whitney U test. All statistical analyses were conducted using the open-source R software (V.3.5.2). For meta-analysis of microbial pathways, we employed the *MMUPHin* (V.1.16.0) method,[48] which accounts for batch effects and cohort heterogeneity. To identify SVs differentially distributed between ASD and neurotypical children, we used a linear mixed-effects model implemented in the *nlme* R package (V.3.1–163). The enrichment of d-SVs in ASD compared with neurotypical children was evaluated using Fisher's exact test. All statistical tests were considered significant at $p<0.05$. To address the issue of multiple testing and reduce the risk of false positives, p values were adjusted using the Benjamini-Hochberg FDR correction.

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Contributors WL designed the study and drafted the manuscript. WL performed bioinformatic analysis. WL and YL collected the samples. SCN and FKLC provided samples. SCN, FKLC and JJYS commented on the study. JY designed, supervised the study and revised the manuscript. JY is the guarantor of this work.

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Patient consent for publication Not applicable.

Ethics approval This study involves human participants and the study protocol was performed in compliance with the Declaration of Helsinki and approved by the Clinical Research Ethics Committee of The Chinese University of Hong Kong (CREC Ref. No.: 2014.026 and 2016.607). Participants gave informed consent to participate in the study before taking part.

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Data availability statement Data are available in a public, open access repository. Raw metagenomics sequencing data of the seven datasets are publicly available from the European Nucleotide Archive (https://www.ebi.ac.uk/ena/browser/home) or Genome Sequence Archive (https://ngdc.cncb.ac.cn/gsa/) via accession numbers (ERP113632, PRJEB23052, PRJNA516054, PRJNA451479, PRJNA782533, CRA004105, PRJEB60702 and CRA005819).

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